

Lab 04: VPC Network Peering

Cloud Security Fundamentals | GSP193

Overview

Set up VPC Network Peering between two projects to enable private connectivity across VPC networks. VPC peering allows internal IP communication between networks without using external IPs or VPNs.

VPC Peering vs Other Options

Feature	VPC Peering	External IPs/VPN
Latency	Lower (private)	Higher (public)
Security	No public exposure	Exposed to internet
Cost	No egress costs	Egress costs apply

Task 1 - Create Custom Networks in Both Projects

Created custom VPC networks in Project A and Project B with VMs and firewall rules.

Project A:

```
gcloud compute networks create network-a --subnet-mode custom
gcloud compute networks subnets create network-a-subnet --network network-a --range 10.0.0.0/16 --region [REGION]
gcloud compute instances create vm-a --zone [ZONE] --network network-a --subnet network-a-subnet --machine-type e2-small
gcloud compute firewall-rules create network-a-fw --network network-a --allow tcp:22,icmp
```

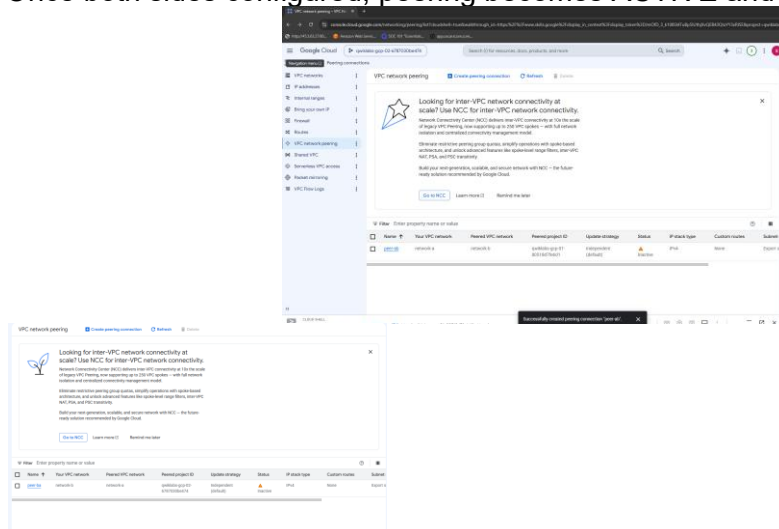
Project B (same pattern with network-b, 10.8.0.0/16):

Task 2 - Set Up VPC Network Peering

Both projects must configure peering independently. Peering stays INACTIVE until both sides are configured.

- Project A: Created peer-ab peering from network-a to network-b in Project B
- Project B: Created peer-ba peering from network-b to network-a in Project A

Once both sides configured, peering becomes ACTIVE and routes are exchanged



Task 3 - Test Connectivity

SSH'd into vm-b and pinged vm-a's internal IP to verify peering is working.

```
ping -c 5 [INTERNAL IP OF VM A]
```

```
SSH-in-browser
linux vm-b 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-01-55c96268f1c6'.
student-01-55c96268f1c6@vm-b:~$ 10.0.0.2
-bash: 10.0.0.2: command not found
student-01-55c96268f1c6@vm-b:~$ ping -c 5 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=36.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=35.6 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=35.3 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=39.5 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=34.1 ms
--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 34.071/35.464/36.828/0.876 ms
student-01-55c96268f1c6@vm-b:~$
```

What I Learned

- VPC peering enables private internal IP communication across projects and organizations
- Both sides must independently configure peering before it becomes ACTIVE
- Peered networks share routes automatically once the connection is active
- Peering provides lower latency and cost than using external IPs or VPNs
- CIDR ranges of peered networks must not overlap

Resources

GCP VPC Peering Docs: <https://cloud.google.com/vpc/docs/vpc-peering>
Lab ID: GSP193